Achernar

Commercial Strategy, Market Position and Revenue Model

Autonomy-First Launch Path

Ibrahim Shahid
Lead Architect

7 August 2026

This document accompanies `achernar_v4.pdf`. Market prices and unit counts in Section 2 are drawn from public sources as of August 2026 and are cited. All revenue figures in Sections 4–6 are **modelled outputs from stated assumptions**, not forecasts. The assumptions — not the totals — are the substance of this document; each is listed in Appendix A so it can be attacked individually.

Contents

1 Executive Summary **2**
1.1 Headline numbers . . . 2
1.2 The honest shape of the opportunity . . . 2

2 Market Landscape **2**
2.1 Price tiers (verified, August 2026) . . . 3
2.2 The white space . . . 3
2.3 Two structural openings . . . 3

3 Repository State **4**
3.1 Built — 19,626 LOC Rust . . . 4
3.2 The single blocker: no trained models, no data . . . 4
3.3 Missing, on the critical path . . . 5

4 Revenue Model **5**
4.1 Method . . . 5
4.2 Segments and total addressable market . . . 5
4.3 Year-5 steady-state scenarios . . . 5
4.4 Five-year ramp (base case) . . . 5
4.5 Breakeven thresholds . . . 6

5 Maximising Revenue: Strategies Ranked **6**
5.1 1. Per-scope recurring revenue from hosting operators . . . 6
5.2 2. OEM licensing of the autonomy stack . . . 6
5.3 3. Smart-telescope attach — the only path with unit scale . . . 6
5.4 4. Vertical integration — operate, do not licence . . . 7
5.5 5. Institutional site licences . . . 7
5.6 6. Prosumer — deliberately not a revenue line . . . 7

6 The Maximum-Money Thesis **7**
6.1 The binding constraint is market size, not share . . . 7
6.2 Three market-expanding moves . . . 7
6.3 The realistic liquidity event . . . 8

7 What Kills Each Path **9**

8 Implementation Sequence **9**
8.1 Commercial milestones mapped to technical ones . . . 10

A Assumptions Register **11**

B Sources **11**

1 Executive Summary

Achernar's commercial value does not lie in being a better capture application. That market is served by mature free software and cannot be won on price. It lies in **Section 7 and Section 8 of v4** — learned closed-loop control with calibrated uncertainty, bounded by a deterministic safety supervisor, and validated through shadow mode against explicit promotion criteria, fed by a single shared perception bus.

No competitor ships this. Every incumbent automates through deterministic scripting and heuristics.

The thesis in one line: do not sell imaging software; sell *unattended operation*, priced against the operator labour it removes.

1.1 Headline numbers

Quantity	Modelled value
Total addressable market (annual, all segments)	$8,490,000
Year-5 ARR — conservative	$126,700
Year-5 ARR — base	$708,800
Year-5 ARR — aggressive	$1,853,000
Cumulative 5-year revenue — base	$1,630,240
Plausible acquisition value (base, 3–8× ARR)	$2.1M–$5.7M
Scopes needed to sustain a solo founder	167
OEM deals needed to sustain founder + 1 engineer	1

1.2 The honest shape of the opportunity

This is a **small-market business**. The entire annual TAM across every segment is under $9M. Achernar is not a venture-scale company and should not be planned as one. It is, on the base case, a sustainable one-to-three person software business with a realistic $2M–$6M acquisition outcome, and on the aggressive case a $5M–$15M one.

The constraint is not execution and it is not the technology. It is that there are only a few thousand robotic telescopes on Earth that anyone pays to operate. Section 6 addresses the only lever that changes this: strategies that *expand* the market rather than dividing the existing one.

2 Market Landscape

2.1 Price tiers (verified, August 2026)

Tier	Product	Price	Note
Free / OSS	NINA, Ekos/KStars, PHD2, Siril, ASTAP	$0	Mature; broad hardware coverage
Hardware bundle	ZWO ASIAIR Plus	$299 one-time	**ZWO gear only**
Prosumer	Voyager	€129 + €29/yr	Markets "AI driven automation"; DragScript / RoboTarget
Prosumer	Sequence Generator Pro	Paid licence	Scripted sequencing
Institutional	**ACP Expert**	**$1,495**	Dispatch scheduler, PinPoint, remote observatory
Institutional	TheSkyX Professional	Paid licence	TPoint, dome and camera control
Service	iTelescope membership	$40–$160/mo	End-user access to hosted rigs
Service	DSP Remote hosting	from $300/mo	Per-rig hosting fee

Smart telescope hardware market: **$221M (2025) → $360M (2032)**, 7.28% CAGR. The ZWO Seestar S50 alone sold **over 50,000 units** before its early-2026 discontinuation, replaced by the S50 Pro at $599–$799.

2.2 The white space

Every product in the table above automates through **scripts and rules**. Voyager's DragScript is genuinely sophisticated, but it is deterministic scripting with heuristic triggers, not learned control. Nothing on the market ships:

- learned closed-loop control with **calibrated uncertainty**;
- a **deterministic safety supervisor** bounding every model action;
- **shadow-mode evaluation** with explicit promotion criteria;
- **cross-subsystem shared perception** — one vision loop feeding focus, filter wheel and mount from a single observation.

v4 §7 and §8 specify all four. This is a real gap in the market, not a positioning exercise.

2.3 Two structural openings

Opening 1 — ACP Expert at $1,495 is the price anchor

ACP Expert sells at roughly ten times the prosumer tier and it sells on exactly one property: unattended operation. Institutions demonstrably pay that figure for automation built on two-decade-old dispatch scheduling. This validates the pricing thesis for an autonomy layer far more strongly than any hobbyist subscription analysis could.

Opening 2 — ASIAIR is ZWO-only

Achernar's INDI / ASCOM / direct-USB abstraction is vendor-neutral, which is the one property ZWO structurally cannot copy without cannibalising camera sales.

Caveat. `hardware/indi_client.rs` is currently a 12-line stub. The moat is specified and not yet built. Until it is, the vendor-neutrality argument is a claim, not a capability.

3 Repository State

Verified against code on 7 August 2026, not inferred from documentation.

3.1 Built — 19,626 LOC Rust

Module	LOC	State
`hardware/`	3,547	camera, focuser, mount, filter wheel, OLS FFI, gphoto2 DSLR, raw decode, discovery
`vision/`	3,115	**vision loop at 1–5 Hz, SharedObservation bus (§8.1 complete)**, classical HFR / star detection, background, trailing, FQE runner
`api/`	3,021	camera, focus, focuser, mount, websocket
`focus/`	2,553	agent state machine, classical solver, backlash, drift, features, policy
`storage/`	1,944	15 repositories, migrations 001–005
`mount/`	693	guiding bandit, dither policy, plate-solve trigger, pointing model, safety
`filter_wheel/`	466	SNR scorer, bandit, planner, controller, safety
`control/`	436	**safety guard, action arbiter, control supervisor, ownership**
`models/`	374	registry with SHA-256 and `runtime_eligible` gating

Plus `ml/` at 4,501 LOC (excluding virtualenv): dataset builders, FQE and offset training, ONNX export, evaluation, replay harness, synthetic PSF, AstroM3 loader with checkpoints downloaded.

The v4 §7/§8 control scaffolding is **essentially complete**. This is the commercially differentiating layer, and it exists.

3.2 The single blocker: no trained models, no data

- `models/focus_quality/1.0.0/metadata.json` declares `"trained": false`, `"weights": "random"`, `"runtime_eligible": false`, `"promotion_status": "never -- integration placeholder only"`.
- `focus_offset/`, `cloud_classifier/` and `trailing_detector/` carry **no `training_info` block at all**, so they fail `is_runtime_eligible()` (`models/model_registry.rs:55`) as well.
- `frames` table: **0 rows**. `focus_samples`: 7. `ml/data/`: a README with four unchecked TODOs.

The architecture is correctly refusing to run itself. Both `model_registry.rs:55` and `vision/fqe_runner.rs:325` gate on `runtime_eligible`, so every model path stays dark until real weights exist. That is the design working as intended — but it means **the data flywheel has never turned once**.

3.3 Missing, on the critical path

- `processing/` is **58 LOC total**. `debayer.rs`, `calibration.rs` and `stacking.rs` are four-line `pub struct` declarations with TODOs. `plate_solver.rs` always returns `Unsupported`.
- `hardware/indi_client.rs` — 12-line stub (the moat, above).
- `ai/` (18 LOC) is dead code, superseded by `vision/`. Delete it.

4 Revenue Model

4.1 Method

Bottom-up by segment: universe size × annual revenue per unit × penetration. Universe sizes are grounded in the public data of Section 2 where it exists and estimated where it does not; every estimate is flagged in Appendix A. Penetration rates are judgement.

$$\mathrm{ARR}_{\mathrm{segment}} = N_{\mathrm{universe}} \times p_{\mathrm{penetration}} \times \mathrm{ARPU}_{\mathrm{annual}}$$

4.2 Segments and total addressable market

Segment	Universe	ARPU/yr	TAM/yr
Prosumer autonomy licence	20,000	$110	$2,200,000
Hosting operator (per scope)	3,500	$540	$1,890,000
Institutional site licence	800	$3,000	$2,400,000
OEM autonomy stack	8	$250,000	$2,000,000
Total			**$8,490,000**

ARPU derivations. Prosumer: $149 perpetual amortised plus $39/yr renewal, priced just above Voyager. Hosting: $45/scope/month. Institutional: anchored on ACP Expert's $1,495 perpetual, converted to a $3,000/yr site licence covering multiple instruments. OEM: flat licence plus royalty.

4.3 Year-5 steady-state scenarios

Segment	Conservative	Base	Aggressive
Prosumer autonomy licence	$22,000	$88,000	$176,000
Hosting operator (per scope)	$56,700	$226,800	$567,000
Institutional site licence	$48,000	$144,000	$360,000
OEM autonomy stack	$0	$250,000	$750,000
Total ARR	**$126,700**	**$708,800**	**$1,853,000**
Customers / units	321	1,269	2,773

Penetration: prosumer 1/4/8%; hosting 3/12/30%; institutional 2/6/15%; OEM 0/1/3 deals.

4.4 Five-year ramp (base case)

Year	ARR	Cumulative	Milestone
Y1	$21,264	$21,264	Shadow-mode validation; design partners; no revenue at scale
Y2	$106,320	$127,584	First paid hosting sites; prosumer beta
Y3	$283,520	$411,104	Institutional entry; prosumer GA
Y4	$510,336	$921,440	OEM deal lands; hosting expands
Y5	$708,800	$1,630,240	Steady state

4.5 Breakeven thresholds

The most useful table in this document, because it converts strategy into a countable target.

Cost base	Burn/yr	Scopes	Sites	Prosumers	OEM deals
Solo founder, full-time	$90,000	167	30	818	0.4
Founder + 1 engineer	$240,000	444	80	2,182	1.0
Team of 4	$620,000	1,148	207	5,636	2.5

167 scopes. That is the entire requirement to make Achernar a full-time job at $45/scope/month. Four mid-size hosting sites. This is the single most important number in the document — it is small enough to be reachable and it does not depend on the prosumer market at all.

5 Maximising Revenue: Strategies Ranked

Ranked by expected value, which is not the same as ranked by ceiling.

5.1 1. Per-scope recurring revenue from hosting operators

Best revenue per unit of effort in the near term.

Commercial hosting sites charge from $300/month per rig. At $45/scope/month you are asking for 15% of their gross on a line item that removes operator labour — an easy internal justification. Sales cycle is short because the buyer is an owner-operator, not a procurement committee.

- **Why it wins:** recurring, scales with fleet size, and the customer's own economics do the selling.
- **Target:** 30–60 commercial facilities worldwide. Perhaps 20 conversations to first contract.
- **Requirement:** INDI client (they run heterogeneous gear), and demonstrated fallback and clamp rates in shadow mode.

5.2 2. OEM licensing of the autonomy stack

Highest ceiling per unit of effort.

Every non-ZWO vendor — Player One, QHYCCD, ToupTek, Pegasus Astro, PrimaLuceLab, Celestron, Sky-Watcher — needs software parity against ASIAIR and none of them want to build a nine-model control stack. You sell ONNX bundles, the safety supervisor and an SDK; they sell hardware.

One deal at $250,000/yr sustains a founder and an engineer outright. That is the highest-leverage single transaction available.

5.3 3. Smart-telescope attach — the only path with unit scale

The Seestar S50 sold **over 50,000 units**. That is the only genuinely large number anywhere in this analysis. Powering a credible number-two vendor:

Model	15,000 units	30,000 units
One-time royalty @ $15/unit	$225,000	$450,000
Subscription attach @ $5/mo, 12% take	$108,000/yr	$216,000/yr
Subscription attach @ $5/mo, 25% take	$225,000/yr	$450,000/yr

5.4 4. Vertical integration — operate, do not licence

Hosting revenue per rig per month	$300
Rigs at one mid-size site	40
Site gross revenue per year	$144,000
Same site licensed at $45/scope/month	$21,600/yr
Revenue multiple of owning vs. licensing	**6.7×**

Owning the site captures 6.7× the revenue of licensing software to it, and the autonomy stack is precisely what lets one operator run a larger fleet. The cost is capital, land, weather risk and operations — a fundamentally different company. Correct as a year-4+ move, wrong as a starting point.

5.5 5. Institutional site licences

ACP-anchored at $3,000/yr. Slow (6–18 month cycles, grant budgets) but extremely sticky and reference-generating. 30 sites sustains a solo founder. Pursue in parallel with hosting; do not make it the first market, because the cycle length will starve you.

5.6 6. Prosumer — deliberately not a revenue line

Recommendation: make the prosumer tier free.
At $88,000 base-case ARR it is the smallest segment in the model, it carries the highest support burden per dollar, and it competes directly with free incumbents. Its actual value is as **the data flywheel** — the diverse rigs, skies and failure modes that every model in §7.6 needs — and as the credibility that makes the OEM and institutional conversations possible.
Charging $110/yr for it trades a $2M strategic asset for a $88k revenue line.

6 The Maximum-Money Thesis

6.1 The binding constraint is market size, not share

Every strategy in Section 5 divides a TAM under $9M. Even perfect execution across all four segments yields under $2M ARR. Optimising pricing or penetration inside that box cannot produce a large outcome.

The only strategies that materially change the ceiling are those that **expand the number of telescopes operating unattended**, rather than capturing a larger share of the few thousand that already do.

6.2 Three market-expanding moves

(a) Drive the cost of unattended operation below the hosting price

Remote hosting exists because operating a remote rig requires a human. If Achernar's autonomy stack makes a $2,000 backyard rig genuinely unattended — roll-off roof, weather abort, automatic recovery, no laptop, no babysitting — then the addressable population is not 3,500 hosted scopes, it is every serious imager with a permanent pier. That is a two-order-of-magnitude expansion of the same product.

(b) Own the smart-telescope autonomy layer

50,000 Seestar units in roughly two years is the only real volume in this market, and that volume exists because the product removed operation from the user entirely. The second generation of smart telescopes will compete on capability under autonomy — longer integrations, mosaics, multi-night projects, transient response. That is exactly the §7 stack. Being the layer inside three vendors' products is worth more than being an application.

(c) Sell the fleet layer, then become the fleet

Land hosting operators as customers first (Section 5.1), learn the operational economics from inside, then vertically integrate at 6.7× the revenue per site. The software business is the reconnaissance for the operating business.

6.3 The realistic liquidity event

Niche vertical software in hardware-adjacent markets clears at roughly 3–6× ARR; a strategic buyer defending a platform position pays more.

ARR at exit	**3×**	**5×**	**8× (strategic)**
$250,000	$750,000	$1,250,000	$2,000,000
$709,000 (base)	$2,127,000	$3,545,000	$5,672,000
$1,853,000 (aggressive)	$5,559,000	$9,265,000	$14,824,000
$3,000,000	$9,000,000	$15,000,000	$24,000,000

Plausible acquirers: ZWO (defending ASIAIR against a vendor-neutral rival), Celestron or Sky-Watcher (needing a Seestar answer), Software Bisque or Diffraction Limited (modernising an ageing institutional stack), or a hosting operator buying its own supply chain.

Accumulating ARR is probably not how this returns the most money. A strategic acquisition at 5–8×, driven by a buyer who needs the autonomy stack to defend a hardware position, is both faster and larger than organic growth inside a $9M TAM. Build accordingly: prioritise the *demonstrable, differentiated capability* and the reference customers that prove it, over revenue maximisation per seat.

7 What Kills Each Path

Path	Primary risk	Mitigation
Hosting operators	One mount crash destroys the reference account and the reputation with it	Safety supervisor is already built; never ship above supervised-auto without shadow-mode promotion evidence
OEM licensing	ZWO bundles a worse autonomy layer for free	Vendor neutrality — build the INDI client; ZWO cannot follow without cannibalising camera sales
Institutional	6–18 month cycles starve a solo operation	Never make it the first market; run in parallel with hosting revenue
Prosumer	Competes with free; support burden exceeds revenue	Make it free by design; harvest data instead
Vertical integration	Capital, weather, land, operations — a different company	Year-4+ move only, funded from software revenue
All paths	MIT licence on a system that moves hardware	Relicense the control and ML layers before any commercial release

Licensing. The README declares MIT. A permissive licence on a closed-loop control system that physically drives a mount is both a commercial error (anyone may fork the moat) and a liability exposure. Resolve before any external distribution.

8 Implementation Sequence

Commercial strategy is downstream of one fact from Section 3: **the `frames` table has zero rows**. Nothing in this document is reachable until the flywheel turns once.

1. **Compile.** `HANDOFF-next-session.md` records that the capture rework was never compiled — no Rust toolchain in the sandbox. Run `cargo build && cargo test` locally. Expect signature mismatches. Nothing below is real until this passes.
2. **Turn the flywheel once.** Capture end-to-end until `frames` receives its first row.
3. **Debayer and calibration** (`processing/`, currently 8 LOC of TODO). FQE consumes 64×64 crops; without debayering, an OSC or DSLR frame yields HFR measured across a Bayer mosaic — corrupt labels. This precedes any training.
4. **Focus sweep harness → first dataset.** Automated ramp across focuser positions, capture at each, classical HFR as label (§9.1 bootstrapping). One clear night yields a few thousand labelled crops. `ml/data_pipeline/build_focus_quality_dataset.py` already exists to consume it.
5. **Train FQE for real.** The chain already exists end to end:

```
ml/train/train_focus_quality.py
  -> ml/train/calibrate_uncertainty.py
  -> ml/export/export_quality_onnx.py
  -> ml/export/write_model_metadata.py
```

Then flip metadata to `trained: true`, `runtime_eligible: true`, `promotion_status: "shadow"`. That single change lights up the shadow path already wired into the vision loop at line 120.

6. **Shadow-mode evaluation.** `ml/eval/replay_offset_policy.py` exists. Compare FQE against classical HFR over logged sessions; apply §7.10 promotion criteria.

7. **INDI client.** The vendor-neutrality moat, and the prerequisite for every hosting-operator conversation in Section 5.1.

Why FQE first. Not because focus matters most, but because it is the *only* model where classical bootstrapping (§9.1) produces labels for free. Every other model in §7.6 requires human labelling or nights of policy rollout. Close one full lap — train, export, shadow, promote — and every remaining model is the same lap with a different label source. Prove the lap once, carefully.

8.1 Commercial milestones mapped to technical ones

Technical milestone	Commercial unlock
First FQE promotion out of shadow mode	Evidence that the promotion machinery works — the core claim of the pitch
Fallback and clamp rates logged over 20+ nights	The only data a hosting operator will actually ask for
INDI client operational	Hosting and institutional conversations become possible at all
Three models promoted, two rigs	First paid design partner
Unattended multi-night session, zero interventions	The demo that closes OEM conversations

A Assumptions Register

Each row is a load-bearing assumption. Attack these rather than the totals.

Assumption	Value	Basis / confidence
Paid-automation user base	20,000	Estimated. Voyager + SGPro class installed base. No public figure; could be 10k or 40k.
Commercially hosted rigs worldwide	3,500	Estimated from 30–60 known facilities at 20–100 rigs each.
Small-aperture robotic + teaching observatories	800	Estimated from 350+ US and 600+ non-US professional observatories, small-aperture subset.
Non-ZWO OEM candidates	8	High confidence. Enumerable vendor list.
Hosting ARPU	$45/scope/mo	Judgement. 15% of the $300/mo hosting fee. Untested.
Institutional ARPU	$3,000/yr	Anchored on verified ACP Expert $1,495 perpetual.
OEM deal value	$250,000/yr	Low confidence. No comparables in this market.
Hosting fee per rig	$300/mo	Verified — DSP Remote published pricing.
Seestar S50 units	50,000+	Verified from multiple retail and review sources.
Smart telescope market	$221M → $360M	Verified — published market research, 7.28% CAGR.
Penetration rates	1–30%	Judgement only. The weakest input in the model.
Exit multiple	3–8× ARR	Sector-typical for niche vertical software.

Sensitivity. Base-case ARR is dominated by two inputs: the OEM deal ($250k of $709k, 35%) and hosting penetration. If no OEM deal lands, base case falls to $459k. If hosting penetration is 5% rather than 12%, it falls a further $132k. The model is therefore best read as: **$300k–$700k is the realistic band**, with everything above it contingent on a single OEM relationship.

B Sources

- ACP Expert pricing — https://acpx.dc3.com/ and OPT Telescopes
- Voyager pricing — https://software.starkeeper.it/voyager-advanced/
- DSP Remote hosting — https://www.dspremote.com/plans-and-pricing/
- iTelescope membership — https://www.itelescope.net/
- Smart telescope market size — https://www.verifiedmarketreports.com/product/smart-telescope-market/
- Seestar S50 units and discontinuation — https://www.telescopeadvisor.com/seestar-s50-discontinued-what-to-buy-instead/
- Observatory counts — https://www.go-astronomy.com/observatories.htm
- Las Cumbres network — https://lco.global/about/
- NINA — https://nighttime-imaging.eu/